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Changes of the geomagnetic field vector obtained from lava sequences on the island of Vulcano (Aeolian Islands, Sicily)

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Abstract

A combined geochronologic (K/Ar) and palaeomagnetic study has been conducted on five volcanic sections (70 flows) on the island of Vulcano. The sections span the interval from 15 ± 2 to 135 ± 4 ka, with large hiatuses. Rock magnetic investigations indicate that the natural remanent magnetisation (NRM) is carried by titanomagnetites in the four oldest sections, with a contribution from high coercivity magnetic minerals in the youngest section. Palaeomagnetic directions obtained by thermal demagnetisation document a large magnetic excursion, at around 110 ka, with virtual geomagnetic poles (VGPs) that lie over eastern Siberia. These directions may represent a volcanic record of the termination of the Blake event. Only about 15% of the samples (from 19 flows) proved suitable for palaeointensity (Thellier and Thellier) determinations. These measurements indicate that the geomagnetic field has varied at Vulcano between 16 and 53 μT in the explored time interval. When combined with previous results obtained at Mt. Etna for the period from 140 to 60 ka (Tric et al., 1994, *Phys. Earth Planet. Inter.*, 85: 113–129), these results document a broad low at around 115 ka. The palaeointensity values from Vulcano and Etna are consistent with global values, suggesting the presence of a dominant dipole-field contribution. These values, on the other hand, are significantly smaller than those recently obtained from La Réunion in the southern hemisphere. This suggests that large, long-lived, non-dipolar components of the geomagnetic field may be present at La Réunion in the interval from 130 to 90 ka. These observations are consistent with the fact that the VGP angular dispersion, calculated from non-excursional data, is significantly smaller at Vulcano than at La Réunion.

1. Introduction

Magnetic observatory measurements of the geomagnetic field vector over the last 4 centuries have provided a detailed picture of the historical and present characteristics of the field and they have led to the development of models that have improved our understanding of the geodynamo. However, longer-term appraisal of the processes that generate the geomagnetic field can only be obtained from palaeomagnetic observations. Several studies, for instance, indicate that there are significant non-dipole field features which are detectable in the time-averaged field as recorded by lavas over the last 5 Myr (Johnson and Constable, 1995). Their persistence suggests that the mechanisms driving the geodynamo

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might be controlled by long-lived conditions at the core–mantle boundary (Gubbins and Kelly, 1993).

Contrary to determinations of relative field intensity changes which can be obtained from deep-sea sediments around the world, absolute values of the geomagnetic field intensity can only be obtained from sequences of lava flows. These studies are hampered by the limited availability of long sequences of flows and often also by our inability to date lava flows with sufficient accuracy to compare results from distant sites. In addition, all of the laboratory methods that are used to retrieve the intensity of the geomagnetic field from lavas involve multiple heating which may change the mineralogical characteristics of the samples during the experiments, thus preventing reliable palaeointensity determination from a significant fraction of samples collected for such studies. Consequently, the spatial and temporal distribution of the measurements of the geomagnetic field vector are far from being exhaustive and homogeneously distributed. Additional studies of lava sequences, well distributed in space and with accurate time control, are thus needed to improve our knowledge of geodynamo processes.

We report new results obtained from sequences of lava flows on the island of Vulcano, southern Italy. Radiometric dating indicates that these sequences comprise two time intervals, the oldest of which partly overlaps with that covered by a sequence at Réunion Island from which we have obtained palaeointensity results (Raïs et al., 1996). These new results, combined with those from Mount Etna (Tric et al., 1994), allow a detailed comparison of geomagnetic field intensity determinations from the northern and southern hemispheres.

2. Geological setting and sampling

The Aeolian archipelago, which comprises seven volcanic islands, is located in the Tyrrhenian Sea near the northeastern coast of Sicily (Fig. 1). The island of Vulcano, at the southern termination of the archipelago, comprises five main structural units (Fig. 1), which delineate the spatial and temporal evolution of the area (Keller, 1980). The oldest unit, Vulcano Vecchio (dated around 110 ± 6 ka) comprises trachibasaltic to trachyandesitic lava flows and

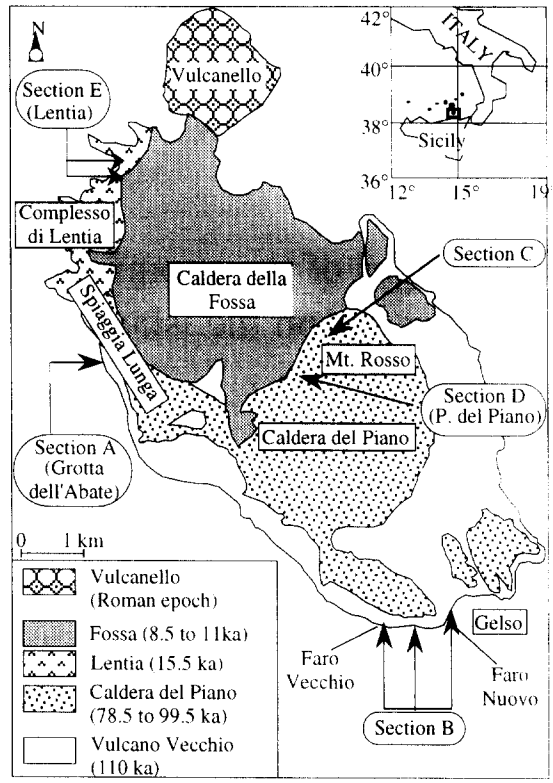


Fig. 1. Simplified geological map of Vulcano with location of the studied sections.

pyroclastic deposits. This volcanic stage was followed by collapse of the stratovolcano in the central part, which gave rise to a second unit, the Caldera del Piano, which is a circular depression filled with lava flows and pyroclastic and scoria deposits. The Caldera was effusively active during the interval from 99.5 to 78.5 ka. In the northwestern part of the island, the Complesso di Lentia comprises rhyolitic units dated around 16 ka. The Caldera della Fossa in the northern part, where volcanic activity started around 11 ka, is the presently active part of the island with prominent fumarolic activity. Vulcanello, which comprises leucitic to nephritic lavas, is of Roman age.

The location of the sampled sections is indicated in Fig. 1. Forty-two lava flows were sampled at Section A, from the Vulcano Vecchio unit, which crops out along the beach of the Grotta dell'Abate. Thin scoriacious levels between flows usually make possible the unambiguous identification and lateral

correlation of the stratigraphic succession. However, the section is occasionally covered, and in two cases it proved difficult to assess whether they were double exposures of the same flow. Flows G40 to G13 (the lower and the middle part of the section) consist of K-rich andesite, while flows G12 to G01 (the upper part of the section) consist of tephrite.

Section B, which also belongs to the Vulcano Vecchio unit, is a composite section. The youngest part of the section, (B1), comprises two flows immediately west of the Faro Nuovo and 13 flows which were sampled further west in the second canyon along the beach. The older part of the section, (B2), consists of three flows (VU 37–39), which were sampled near the dykes west of the Faro Vecchio.

The other sections were sampled from more recent units: Section C (five flows, around 98 ka) northwest of Monte Rosso, Section D (five flows, around 78 to 52 ka; Gillot, 1987) in the Passo del Piano and Section E (nine flows, around 20 ka) in La Lentia.

At least seven cores were sampled per flow (590 cores total). The cores were oriented in situ using a magnetic compass and a sun compass. Each core yielded five to six standard palaeomagnetic samples after cutting in the laboratory. Fresh massive samples with different petrologic natures were sampled at different stratigraphic horizons for radiometric dating.

3. Results

3.1. Radiometric dating

Radiometric dates were obtained with an unspiked K–Ar technique which was developed specifically for the detection of very low radiogenic argon percentages (Cassignol and Gillot, 1982; Gillot and Cornette, 1986). Several technical improvements make it possible to obtain negligible drift during the measurements which were made in a semi-static vacuum mode. This allows the $^{40}Ar/^{36}Ar$ ratio of the sample and the laboratory atmospheric $^{40}Ar/^{36}Ar$ ratio to be measured successively in strictly the same working conditions, thus avoiding the need for a spike. The value of the initial Ar isotopic ratio does not need to be exactly known, the only assumption being that the atmospheric ratio at the time of eruption of the flow is the same as today. The ^{40}Ar content can be determined with a precision of 0.4% ($\pm 2\sigma$). This precision and the overall performance of the technique have been established by cross-calibration experiments on reference standard minerals (GL-O: Cassignol and Gillot, 1982; Odin et al., 1982; MMhb-1: Samson and Alexander, 1987; LP-6: Odin et al., 1982; HD-B1: Fuhrmann et al., 1987). Recent reexamination has further improved the accuracy of the cross-calibration (Charbit et al., 1995). This technique has already been successfully used for dating young rocks (Gillot et al., 1982; Garnier et al., 1996; Raïs et al., 1996) and the experimental details have been extensively described (Cassignol and Gillot, 1982; Gillot and Cornette, 1986).

In this work, the samples were crushed and sieved to between 0.25 and 0.125 mm and ultrasonically washed in HNO_3, in order to remove weathered phases. Both K and Ar measurements were performed on the micro crystalline groundmass which is considered to be representative of the phase that crystallised during solidification of the lavas. This phase, which concentrates the K is the most homogeneous phase for dating. Phenocrysts and xenocrysts were removed by using both heavy liquids of appropriate densities and magnetic separations, in order to exclude inherited extraneous Ar from either early crystallisation or from lithic fragments which may have been incorporated from surrounding pre-existing rocks. The absence of such material in the aliquots was carefully checked with an optical microscope. K was analysed by atomic absorption and flame emission spectrophotometry. Ar was extracted from 1-g to 2.5-g groundmass samples by radio frequency heating induction in a high vacuum glass line, and was purified with titanium sponge and Zr–Al getters.

The analytical data and the calculated ages are reported in Table 1, together with previously obtained data (Gillot, 1987). The data obtained here correspond to the mean value of two independent replicate measurements for K and for Ar (the consistency of replicate determinations is a good check of the accuracy of age determinations). Age calculations are based on the following decay and abundance constants: $\lambda\beta^- = 4.962 \times 10^{-10}$ a^{-1}; $\lambda_\varepsilon = 0.581 \times 10^{-10}$ a^{-1}; $40K/K = 1.167$ 10^{-4} mol mol^{-1} (Steiger and Jäger, 1977). Standard errors

Table 1
Radiometric K–Ar dates from the sampled sections on Vulcano (Aeolian Islands, Sicily)

Flow number	K (wt.%)	$^{40}Ar^*$ (%)	$^{40}Ar^*$	Age ($\pm 2\sigma$) ka
Section A (Grotta dell'Abate)				
G04	3.189 ± 0.032	4.231	3.747 ± 0.091	112 ± 3
	3.189 ± 0.032	4.386	3.733 ± 0.087	112 ± 3
G14	3.859 ± 0.039	3.991	4.329 ± 0.111	107 ± 3
	3.859 ± 0.039	4.270	4.433 ± 0.106	110 ± 3
G22	3.914 ± 0.039	8.005	4.495 ± 0.060	110 ± 2
	3.914 ± 0.039	7.995	4.343 ± 0.059	106 ± 2
Section B (Gelso)				
VU20	3.591 ± 0.036	5.676	3.912 ± 0.071	104 ± 2
	3.591 ± 0.036	6.339	4.033 ± 0.066	107 ± 2
VU35	3.428 ± 0.034	7.631	3.597 ± 0.050	100 ± 2
	3.428 ± 0.034	8.923	3.645 ± 0.045	102 ± 2
VU39	2.119 ± 0.021	3.755	2.957 ± 0.081	134 ± 4
	2.119 ± 0.021	4.445	3.007 ± 0.069	136 ± 3
Section C (West Monte Rosso)				
VU19	2.970 ± 0.029	2.818	3.048 ± 0.109	98 ± 4
	2.970 ± 0.029	1.893	3.141 ± 0.167	101 ± 5
Section D (Passo del Piano)				
VU10	1.900 ± 0.020	2.478	0.965 ± 0.039	49 ± 4
	1.900 ± 0.020	2.784	1.083 ± 0.039	55 ± 4
VU14	2.450 ± 0.025	4.965	2.019 ± 0.042	79 ± 2
	2.450 ± 0.025	5.382	1.973 ± 0.038	77 ± 2
Section E (Mt. Lentia)				
VU01	4.230 ± 0.042	2.553	0.702 ± 0.028	16 ± 1
	4.230 ± 0.042	1.587	0.653 ± 0.041	15 ± 1
VU08	2.600 ± 0.026	1.304	0.775 ± 0.059	28 ± 2
	2.600 ± 0.026	1.239	0.672 ± 0.054	25 ± 2

K (wt. %), potassium weight percentage; $^{40}Ar^*$ (%), radiogenic Argon percentage; $^{40}Ar^*$, amount of radiogenic Argon (10^{11} atoms g^{-1}).
Data from VU10 and VU19 are from Gillot (1987).

(95% confidence level) were calculated from 100 measurements for each Ar isotope.

The results indicate that the bottom of Section B (135 ± 4 ka) is older than expected (~ 110 ka). For Sections A and B, age determinations are not precisely consistent with their stratigraphic order, but the discrepancies are within the experimental error of the dating itself. This suggests that the flows in these two sections were extruded rapidly. For Section C, only one age determination was obtained (98 ± 4 ka), therefore estimation of the extrusion rate is not possible. For Sections D and E, previous age determinations combined with those from this study indicate that they span the intervals 78 ± 2 to 54 ± 2 ka and 27 ± 2 to 15 ± 1 ka, respectively.

3.2. Rock magnetic analysis

Rock magnetic investigations were conducted on one or two samples per flow to identify the magnetic minerals that carry the natural remanent magnetisation (NRM). Large fluctuations of the low-field magnetic susceptibility (from 1 to 8×10^{-4} SI (Standard International) within and between flows suggest that the concentration of magnetic minerals is highly variable in the sequences. The largest variations were observed from Sections A and B, while less variation was observed from Sections C, D and E.

Strong field thermomagnetic analyses were carried out with a horizontal Curie balance in an Ar atmosphere to minimise oxidation of the magnetic minerals. Heating and cooling rates are of the order of 8–10°C min^{-1}. The results obtained from the analysis of 109 powdered samples allow discrimination of two main types of behaviour.

Type 1 behaviour, observed for ~ 80% of the samples, is characterised by a regular decay of the induced magnetisation from room temperature to about 300–400°C, followed by a rapid decrease to a Curie temperature (T_c) which varies between 540 and ~ 590°C. For 10% of these samples (Type 1a) the cooling curve is very similar to the heating curve showing that the single ferromagnetic phase present in the samples, probably low Ti-magnetite, is stable upon heating (Fig. 2(a)). For the other samples (Type 1b), the cooling curve is shifted toward lower temperatures with a decrease in the saturation magnetisation, J_s, of 5 to 30% (Fig. 2(b)). This decrease in T_c and J_s could be due to temperature-induced Ti diffusion which leads to an increased Ti-content in the Ti-poor titanomagnetite (Soffel, 1975).

Type 2 behaviour is observed in ~ 15% of the samples from Sections A, B, C and D (Fig. 2(c)). They are characterised by two Curie points, the first of which occurs between 450 and 550°C, while the second is usually close to the T_c of magnetite (580°C). The cooling curve documents the 580°C Curie point and it usually lies above the heating curve. The increase in J_s could reflect exsolution of low Ti-

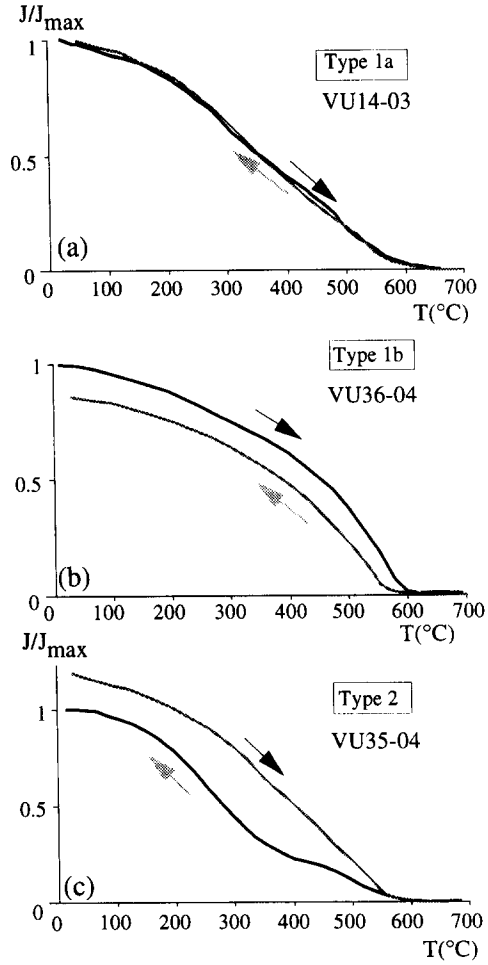


Fig. 2. High field thermomagnetic curves of representative samples (see text for explanation). All curves are normalised to the initial magnetisation.

magnetite, which is known to occur at high temperatures (Ade-Hall et al., 1971).

Very occasionally (< 2% of the samples), a third behaviour is observed. It is characterised by a heating curve with two Curie points. The first one is observed between 450 and 530°C and the second at higher temperatures (610–650°C). The cooling curve shows a Curie point close to the second one and a weaker J_s. This behaviour may be due to the presence of titanomagnetite which partially transforms into hematite with increasing temperature (Ade-Hall et al., 1971).

Magnetic hysteresis measurements of 91 rock fragments (more than one measurement per flow) were performed up to maximum fields of 2 T with an alternating gradient force magnetometer (Fig. 3(a)–(c)). After subtraction of a small paramagnetic component, indicated by the slope at high fields, the saturation magnetisation (M_s), the saturation remanence (M_r), the coercive field (H_c) and the coercivity of remanence (H_{cr}) were calculated. H_{cr} is always lower than 50 mT, except for the lower part of Section E, where values of up to 75 mT were observed (Fig. 3(d)–(f)), which indicates a contribution from high coercivity minerals. Values of the *S*-ratio (Stober and Thompson, 1979), which are typically used to estimate the relative importance of low and high coercivity minerals, are close to unity for all of the sections (Fig. 3(d)) including the upper part of Section E (Fig. 3(e)). For the lower part of Section E, the *S*-ratios are significantly lower than 1 (Fig. 3(f)) confirming that a high coercivity mineral contributes to the magnetisation.

In Sections A, B, C and D, titanomagnetite is the only carrier of the remanent magnetisation. Day diagrams (Day et al., 1977) are characterised by significant scatter (Fig. 4), which may reflect variations in grain size. A significant difference is apparent between Section A, the only section where multi-domain grains are documented, and the other sections where pseudo-single domain grains probably dominate the magnetic mineralogy (Fig. 4). Much of the scatter could, however, be due to variations in the Ti-content (Day et al., 1977) or to oxidation of the different samples.

3.3. Demagnetisation of the NRM

NRM intensities range from 1 to 12 A m^{-1} with a mean intensity between 1.5 and 2 A m^{-1}. Stepwise thermal demagnetisation of the NRM, with a minimum of ten steps, was performed on 240 samples. Most of the samples are characterised by a single stable remanence component that is usually isolated above 150–200°C (Fig. 5).

Normal polarity was observed throughout all the sections. The detailed results obtained from the longest sections (A and B) are reported in Tables 2 and 3. The observed fluctuations of remanence directions are within the range of normal palaeosecular

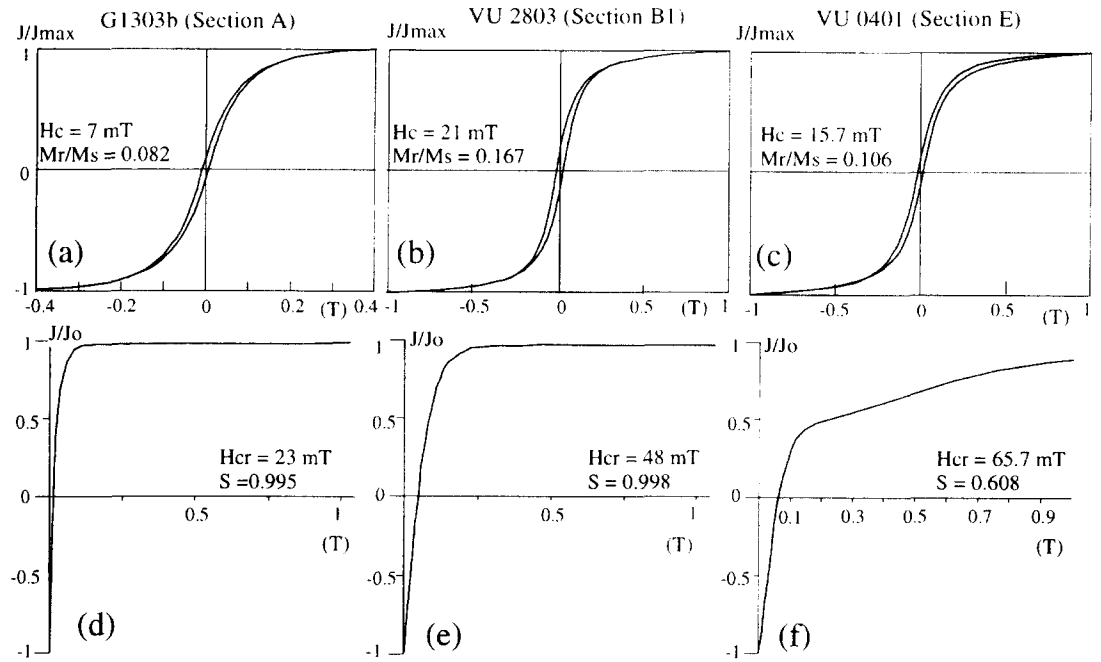


Fig. 3. Typical hysteresis loop and isothermal remanent magnetisation acquisition diagrams for representative samples (see text for explanation).

variation, except for a large deviation from the dipole field direction, which is documented in the lower part of Section A (Table 2 and Fig. 6(a)). The deviation does not correlate with petrological differences (Fig. 6(a)). The corresponding virtual geomagnetic poles (VGPs) cluster over eastern Siberia at a latitude of $\approx 55°$N (Fig. 6(b) and Fig. 7). The cluster of VGPs is recorded by 18 successive lava flows, which have the same K/Ar age, within the experimental accuracy of 3 ka. The high extrusion rate recorded by these data is consistent with volcanological evidence (G. Frazzetta, personal communication, 1995). Thus, it is difficult to estimate the duration of this excursion. Assuming, as a rough estimate, that the flows in Section A were extruded over a 3-ka period and that the extrusion rate was constant, the excursion may have lasted for about 1000 years. This estimate is consistent with the time duration of the clusters of transitional VGPs observed during selected polarity reversals, as suggested by Hoffman (1992, 1994). The geographical position of the cluster is also consistent with one of the cluster points suggested by Hoffman (1992, 1994) and lies on one of the two preferred bands evidenced by Laj et al. (1991, 1992) for reversals recorded in sediments (0–12 Ma).

3.4. Palaeointensity determinations

3.4.1. Sample selection and experimental procedure

Sample selection was made in several steps. First, only samples with a weak secondary overprint were selected (Fig. 5(a)). On this basis, all samples from Section C were rejected. Following other authors (e.g. Coe et al., 1984), we used the thermomagnetic behaviour as a second criterion, and selected samples that are characterised by a single high Curie point and reversible behaviour on cooling (Type 1 behaviour, cf. Fig. 2(a)). Some (less than 20% of the total) Type 2 samples were also considered in time intervals where Type 1 samples were either absent or where they did not yield reliable results (Fig. 2(b) and (c)). A total of 256 samples from Sections A, B, D and E were used for palaeointensity determinations.

The Thellier double-heating method (Thellier and Thellier, 1959) was used for this study. The samples were heated twice (normal and reverse positions)

from room temperature to over 500°C, with at least 11 steps and variable temperature increments of 30°C to 50°C. The furnace has a capacity of 60–80 samples. The heating time ranged from 1 h 45 min to 2 h 30 min, and the samples were allowed to cool naturally overnight. The heating and cooling were done in an Ar atmosphere to minimise oxidation of magnetic minerals. A constant laboratory field of 40 μT, close to the present field intensity at Vulcano, was applied along the z axis of the samples during heating and cooling as suggested by Levi (1975). As magnetochemical changes can occur during heating, the magnetic stability of the samples was checked by remeasuring the partial thermoremanent magnetisation (pTRM) at a given low temperature after heating at a higher temperature. Inconsistent pTRM checks indicate that only part of the NRM can be used to determine the palaeofield intensity.

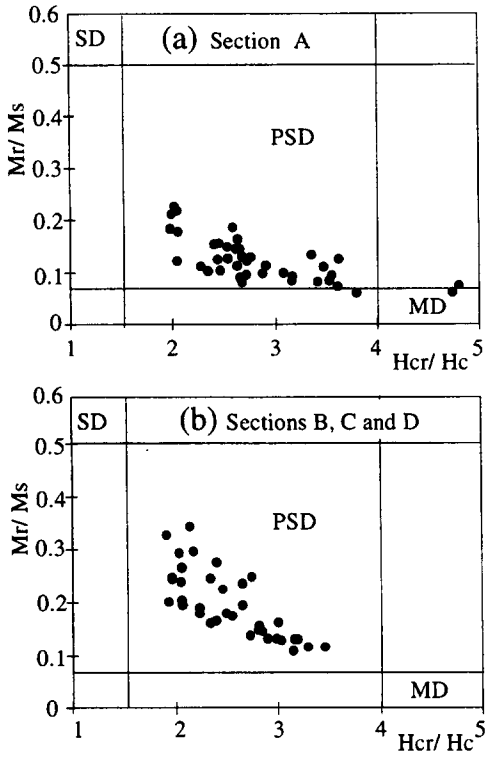


Fig. 4. Plot of M_r/M_s versus H_{cr}/H_c (Day et al., 1977). (a) Distribution of magnetic grain size in the samples from Section A, and (b) for Sections B, C and D. The SD, PSD and MD sectors correspond to those for single-domain, pseudo-single domain and multi-domain magnetite, respectively.

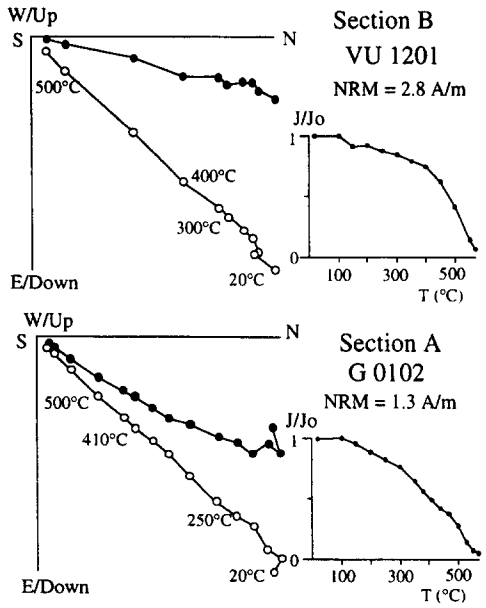


Fig. 5. Typical orthogonal demagnetisation diagrams obtained using thermal demagnetisation. Closed and open symbols correspond to projections onto the horizontal and vertical plane, respectively. NRM decay curves are also indicated.

Examples of the palaeointensity results, represented as NRM/TRM diagrams (Nagata et al., 1963), are given in Figs. 8 and 9. The NRM remaining at each temperature is plotted against the corresponding TRM acquired in the laboratory field at the same temperature. Both NRM and TRM values are normalised to the room temperature NRM. In an ideal diagram, the NRM/TRM points fall on a straight line, the slope of which gives the ratio between the field applied in the laboratory and the Earth's palaeofield. In practice, because of oxidation and magnetochemical changes during heating, only a fraction of the samples usually yield reliable palaeointensity estimates.

3.4.2. Palaeointensity results

The reliability of the palaeointensity determinations was assessed by the: (i) linearity of the NRM–TRM diagrams, (ii) consistency of pTRM checks, (iii) stability of remanence direction, (iv) stability of the low field susceptibility and (v) analytical precision (Coe et al., 1978). Examples of NRM–TRM diagrams for samples that yielded acceptable results

are given in Fig. 8. These diagrams exhibit linear NRM–TRM changes, with positive pTRM checks. The direction of the remanent magnetisation calculated after each step is stable and shows no tendency toward the direction of the laboratory field. Diagrams like those in Fig. 8 were obtained for 40 samples from 19 flows, i.e. about one-fifth of all the samples analysed.

Typical diagrams from non-ideal samples are shown in Fig. 9. Diagrams similar to that for sample G2101 were observed for about one-third of the samples. Such concave-upward diagrams may result from irreversible motion of domain walls in multidomain grains (Levi, 1977). The diagram for sample VU3403, typical of one-quarter of the samples, is characterised by a decrease in TRM acquisition capacity after 300°C and an increase after 450°C. This behaviour may be due to magnetochemical transformations during the multiple heatings. The diagram for sample G0202, typical of one-sixth of the samples, is characterised by a linear NRM–TRM diagram, but the remanence direction moves toward the direction of the field applied during heating and cooling. This behaviour may be due to the transformation of the original magnetic minerals at low temperatures, or to the generation of minerals with high blocking temperatures. The non-ideal NRM–TRM diagrams led to rejection of these samples.

In evaluating the palaeointensity data, one can only use the results obtained below the temperature at which magnetochemical transformations begin to

Table 2
Palaeomagnetic characteristics of Section A (Grotta dell'Abate)

Flows	n/N	D (°)	I (°)	α_{95}	Lat. (°N)	Long. (°E)
G01	1/4	6	37	n.d.	72	177
G02	4/4	355	43	8	76	215
G03	3/4	1	46	3	79	190
G04	4/4	356	47	3	80	215
G06	4/6	358	44	6	78	203
G07	4/4	351	49	5	79	240
G08	3/5	349	54	12	80	267
G08b	3/4	353	50	11	81	236
G09	4/4	355	50	5	82	226
G10	4/4	352	48	3	79	234
G11	4/4	347	51	8	78	258
G13	3/4	345	49	6	75	256
G14	4/4	348	58	15	81	292
G15	3/4	12	60	10	80	84
G16	3/4	347	54	8	79	271
G17	4/4	357	54	5	86	231
G18	4/4	341	49	15	71	257
G20	4/4	352	50	5	80	240
G21	4/4	350	48	5	78	241
G22	4/4	352	49	6	80	237
G23	3/5	6	69	10	75	29
G24	4/5	12	68	10	75	44
C25	3/4	44	56	8	55	94
G26	4/6	44	54	3	55	97
G27	4/4	36	57	7	62	95
C28	4/4	45	47	5	51	106
G29	4/4	35	48	8	60	112
G30	4/4	45	38	4	48	115
G31	3/4	37	55	5	60	98
G32	4/4	39	49	5	57	107
G33	3/4	34	58	19	63	93
G33b	4/4	47	47	7	50	105
G35	3/5	44	52	16	54	100
G36	4/4	47	51	8	51	100
G37	4/4	45	51	7	53	101
G38	4/4	35	50	4	60	108
G39	4/4	45	46	4	51	107
G40	4/4	52	46	3	46	103

n/N, number of successful/attempted determinations; D, I, mean declination and inclination values; α_{95}, 95% confidence angle; n.d., not determined; Lat., Long., latitude and longitude of the VGPs; "b", indicates flows which may be the same as the flow without the "b" label.

Table 3
Palaeomagnetic characteristics of Section B (Gelso)

Flows	n/N	D (°)	I (°)	α_{95}	Lat. (°N)	Long. (°E)
VU20	6/6	3.3	49.0	3.7	81	176
VU23	6/6	357.4	43.1	3.2	76	205
VU24	6/6	359.6	43.1	5.5	77	196
VU25	4/4	6.3	40.7	3.4	74	173
VU26	7/7	356.5	47.2	2.5	79	212
VU27	6/6	5.3	49.0	4.1	80	166
VU28	5/5	347.1	47.8	6.4	76	247
VU32	5/5	359.5	43.6	4.7	77	197
VU33	6/6	3.3	48.5	4.5	81	176
VU34	5/5	4.3	51.5	4.1	83	163
VU35	5/5	359.7	48.8	4.0	81	196
VU36	4/4	353.6	47.0	3.0	78	224
VU36b	6/6	356.0	49.6	3.2	81	218
VU37	8/7	347.5	53.5	2.5	79	267
VU38	7/7	355.7	56.0	6.3	86	262
VU39	6/6	353.0	53.8	4.0	83	253

n/N, number of successful/attempted determinations; D, I, mean declination and inclination values; α_{95}, 95% confidence angle; Lat., Long., latitude and longitude of the VGPs; VU36b may be the same as flow VU36.

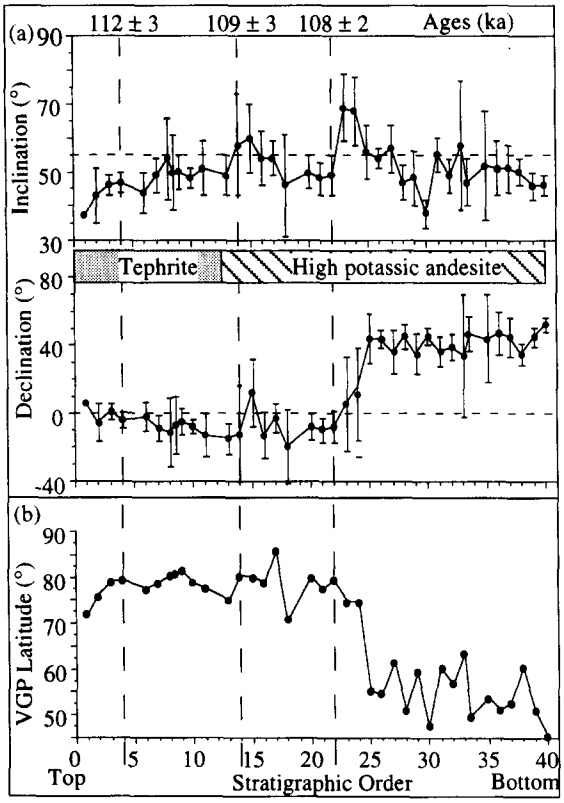


Fig. 6. (a) Changes of the mean declination and inclination from flows of Section A. The horizontal lines correspond to the direction expected at the site for an axial dipole field. (b) Variations of the corresponding VGP latitude.

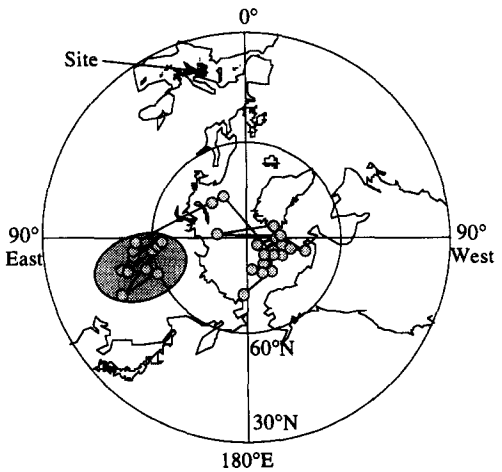


Fig. 7. Distribution of the VGPs obtained from Section A.

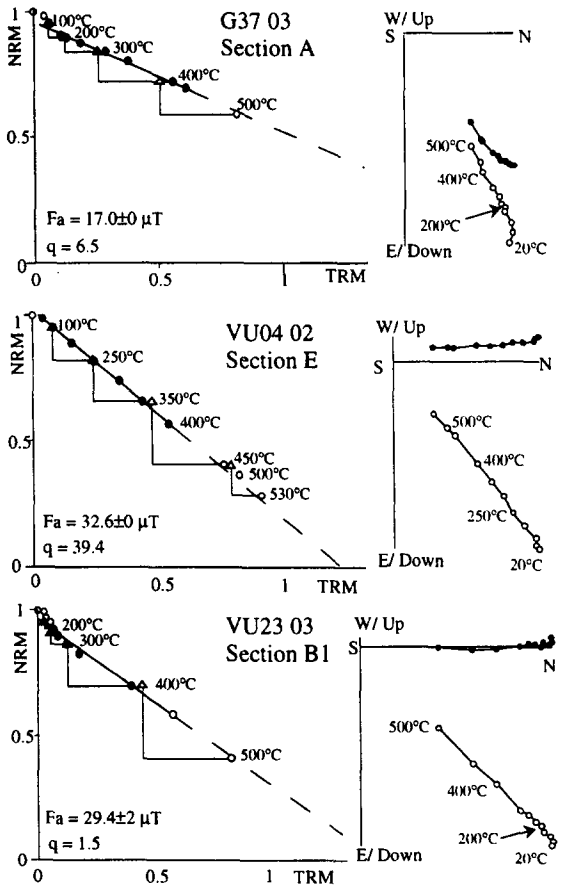


Fig. 8. Examples of NRM–TRM diagrams accepted for palaeointensity determinations together with the corresponding orthogonal projection diagrams (same conventions as in Fig. 5). Triangles correspond to the pTRM checks. Closed symbols on the NRM–TRM plots correspond to data taken into account for the calculation of the regression line. F_a is the calculated palaeointensity with its standard error and q is the quality factor (Coe et al., 1978).

occur, as illustrated by non-linearity in the NRM/TRM curve and by negative pTRM checks. Palaeointensities were then calculated using a least-squares fit to a straight line through the data in the selected temperature interval. The overall quality of the palaeointensity determinations was evaluated using the statistical parameters developed by Coe et al. (1978). Diagrams corresponding to a NRM loss < 15% before the onset of the magnetochemical transformations ($f \leq 15\%$) were not considered reliable,

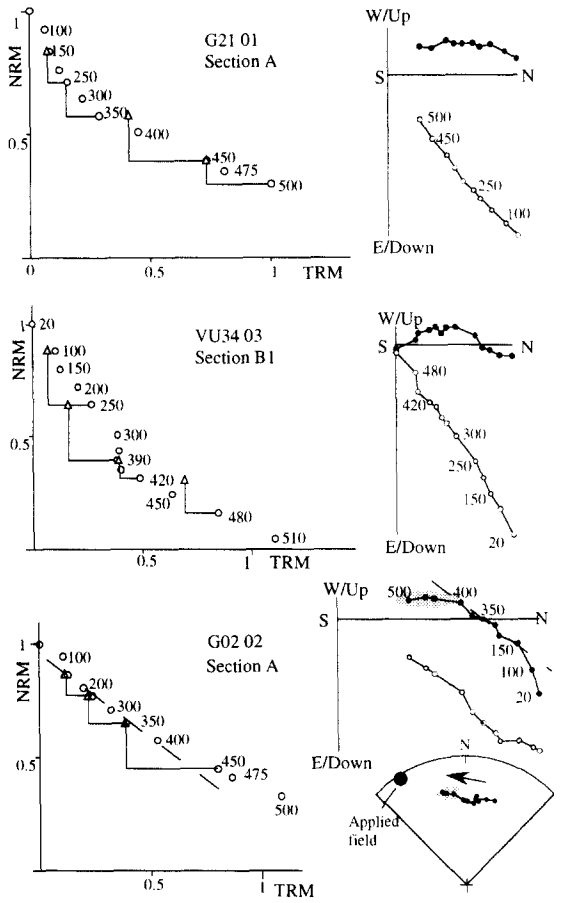


Fig. 9. Examples of rejected (see text for explanation) NRM–TRM diagrams. Conventions are the same as in Fig. 8.

except when the obtained palaeointensity determinations were consistent with more accurate data from the same flow (Table 3).

3.4.3. *Field intensity changes at Vulcano*

The 40 reliable palaeointensity determinations and associated statistical parameters are summarised in Table 4 and in Fig. 10. The left-hand side of Fig. 10 contains the palaeointensity estimates obtained from single samples, while the middle column contains the age determinations from the respective lava flows. The right-hand side of the figure contains the flow-averaged palaeointensity results. The palaeointensity determinations range from 16 to 53 μT, and have average values of 32 $\pm$ 4 μT during the interval from 15 to 27 ka (Section E), of 31 $\pm$ 6 μT at about 60 ka (Section D) and 27 $\pm$ 6 μT in the interval from 104 to 135 ka (Sections A and B). These average values are lower than the present value of 44.5 μT.

4. Discussion and conclusions

Only 40 samples out of 256, from 19 out of 70 flows (15% and 27%, respectively) yielded reliable palaeointensity determinations. This percentage is lower than that usually obtained in similar studies of volcanic formations, and indicates that the units at Vulcano are not, on average, particularly suitable for retrieving the palaeointensity estimates. Nevertheless, the 40 successful determinations, which meet strict selection criteria, can be considered reliable spot readings of geomagnetic field intensity during the explored time interval.

Further evidence of the reliability of these data is available by comparison with results obtained from Mt. Etna (Tric et al., 1994), which is located only about 100 km away from Vulcano (Fig. 11). Because the results from Mt. Etna were obtained from individually dated flows, we have assumed a linear extrusion rate between dated flows at Vulcano to estimate the age of the data points in Sections D and E. In Sections A and B, the extrusion rate is high, and the accuracy of the dating is insufficient to discriminate between flows. We therefore used an

Note to Table 4:
In the first column, we indicate the label of the flow followed by the sample number (flow-sample); T_{min}/T_{max} is the temperature interval over which the Arai diagram was considered reliable; N is the number of points used to determine palaeointensity; D, I are the declination and inclination of the NRM determined between T_{min} and T_{max}; f, g, q are the NRM fraction, gap factor and quality factor relative to the palaeointensity determination (Coe et al., 1978); r is the linear correlation coefficient; $Fa \pm \sigma Fa$ is the palaeointensity estimate for the individual specimens from the Thellier experiment; Fa + s.d. is the simple mean of palaeointensity data from the individual lava flows; Fm + s.d is the mean palaeointensity calculated for each time interval; VADM + s.d. is the virtual axial dipole moment and its standard deviation.

Table 4
Palaeointensity estimates and associated statistical parameters

Flow-sample	T_{min}/T_{max} (°C)	N	D (°)	I (°)	f	g	q	r	$Fa \pm \sigma Fa$ (μT)	$Fa \pm$ s.d. (μT)	$Fm \pm$ s.d. (μT)	VADM $\pm$ s.d. (10^{22} Am^2)
Section E (15–27 ka)												
VU01-02	150/480	8	−3	52	0.32	0.79	3.11	0.980	29.5 ± 2	29.5 ± 2.1	29.5 ± 2.1	5.3 ± 0.4
VU02-00	300/500	6	−9	37	0.34	0.52	5.46	0.996	24.9 ± 0	31.0 ± 6.4	31.0 ± 6.4	5.5 ± 1.2
VU02-02	200/450	6	−7	33	0.07	0.83	0.94	0.981	30.2 ± 2	31.0 ± 6.4	31.0 ± 6.4	5.5 ± 1.2
VU02-03	20/480	12	−8	33	0.41	0.88	3.46	0.922	31.7 ± 4	31.0 ± 6.4	31.0 ± 6.4	5.5 ± 1.2
VU03-02	200/475	7	3	46	0.23	0.73	3.56	0.991	28.4 ± 1	28.5 ± 1.0	28.5 ± 1.0	5.1 ± 0.2
VU04-01	100/390	9	−1	45	0.30	0.83	13.28	0.999	47.4 ± 1	37.5 ± 6.7	37.5 ± 6.7	6.7 ± 1.2
VU04-02	100/400	7	−9	50	0.32	0.78	39.44	1.000	32.6 ± 0	37.5 ± 6.7	37.5 ± 6.7	6.7 ± 1.2
VU04-02	20/450	11	−9	51	0.54	0.88	27.95	0.998	34.2 ± 1	37.5 ± 6.7	37.5 ± 6.7	6.7 ± 1.2
VU04-03	200/420	7	−8	41	0.08	0.78	1.08	0.986	35.6 ± 2	37.5 ± 6.7	37.5 ± 6.7	6.7 ± 1.2
Section D (around 60 ka)												
VU12-02	115/350	6	31	44	0.42	0.78	2.88	0.974	26.5 ± 3	31.0 ± 6.4	31.0 ± 6.4	5.5 ± 1.1
VU12-03	115/350	6	1	20	0.16	0.74	1.47	0.990	35.5 ± 3	31.0 ± 6.4	31.0 ± 6.4	5.5 ± 1.1
Section B1 (around 104 ka)												
VU20-02	150/400	6	−48	56	0.20	0.75	3.75	0.995	36.1 ± 2	34.3 ± 4.0	33.3 ± 5.3	6.0 ± 0.9
VU20-03	150/400	6	2	52	0.22	0.73	3.98	0.995	32.4 ± 1	34.3 ± 4.0	33.3 ± 5.3	6.0 ± 0.9
VU21-02	150/400	6	0	49	0.22	0.73	2.42	0.986	28.1 ± 2	33.8 ± 4.9	33.3 ± 5.3	6.0 ± 0.9
VU21-04	100/420	9	1	49	0.34	0.86	10.59	0.996	36.8 ± 1	33.8 ± 4.9	33.3 ± 5.3	6.0 ± 0.9
VU21-II0	300/500	6	−14	48	0.16	0.74	5.79	0.998	36.4 ± 1	33.8 ± 4.9	33.3 ± 5.3	6.0 ± 0.9
VU23-03	150/450	7	0	43	0.13	0.80	1.47	0.980	29.4 ± 2	29.4 ± 2.1	33.3 ± 5.3	6.0 ± 0.9
VU25-04	20/475	10	6	39	0.32	0.68	8.10	0.996	29.1 ± 1	26.4 ± 5.8	33.3 ± 5.3	6.0 ± 0.9
VU25-03	100/420	9	3	46	0.30	0.86	7.03	0.994	30.4 ± 1	26.4 ± 5.8	33.3 ± 5.3	6.0 ± 0.9
VU25-05	20/400	7	−36	62	0.37	0.78	24.25	1.000	29.7 ± 0	26.4 ± 5.8	33.3 ± 5.3	6.0 ± 0.9
VU26-03	150/300	4	0	46	0.37	0.65	8.84	0.999	28.3 ± 1	35.2 ± 9.8	33.3 ± 5.3	6.0 ± 0.9
VU26-03	150/360	6	−2	45	0.24	0.78	2.15	0.980	42.1 ± 4	35.2 ± 9.8	33.3 ± 5.3	6.0 ± 0.9
VU27-05	20/400	7	−4	56	0.37	0.78	4.22	0.988	52.9 ± 4	43.9 ± 12.7	33.3 ± 5.3	6.0 ± 0.9
VU27-06	20/500	11	−3	44	0.28	0.84	4.84	0.984	34.9 ± 2	43.9 ± 12.7	33.3 ± 5.3	6.0 ± 0.9
VU28-01	300/500	6	2	46	0.48	0.78	3.49	0.952	37.5 ± 4	37.4 ± 0.2	33.3 ± 5.3	6.0 ± 0.9
VU28-01	300/460	5	1	46	0.20	0.71	1.56	0.987	37.2 ± 3	37.4 ± 0.2	33.3 ± 5.3	6.0 ± 0.9
VU33-01	150/400	6	4	47	0.21	0.74	1.84	0.977	26.5 ± 2	26.5 ± 2.2	33.3 ± 5.3	6.0 ± 0.9
VU36-01	200/420	6	−8	47	0.17	0.75	2.29	0.989	24.1 ± 1	31.4 ± 10.3	33.3 ± 5.3	6.0 ± 0.9
VU36-05	180/400	7	−4	43	0.23	0.77	2.46	0.987	38.6 ± 3	31.4 ± 10.3	33.3 ± 5.3	6.0 ± 0.9
Section A (around 119 ka)												
G10-02	150/350	5	−18	50	0.19	0.70	3.50	0.998	36.9 ± 1	36.9 ± 1.3	30.4 ± 9.2	5.4 ± 1.7
G17-01	200/475	7	0	54	0.32	0.75	3.50	0.988	23.8 ± 2	23.8 ± 2.2	30.4 ± 9.2	5.4 ± 1.7
G37-03	200/475	7	53	46	0.19	0.73	5.78	0.998	16.9 ± 1	16.9 ± 1.3	17.3 ± 0.6	3.1 ± 0.1
G39-01	20/475	10	41	46	0.18	0.66	7.91	0.999	18.8 ± 0	17.8 ± 1.7	17.3 ± 0.6	3.1 ± 0.1
G39-05	20/450	9	50	43	0.02	0.76	0.57	0.993	15.7 ± 1	17.8 ± 1.7	17.3 ± 0.6	3.1 ± 0.1
G39-06	20/450	9	49	45	0.12	0.54	1.31	0.987	17.1 ± 1	17.8 ± 1.7	17.3 ± 0.6	3.1 ± 0.1
G39-07	20/475	10	46	46	0.16	0.63	5.04	0.998	19.4 ± 1	17.8 ± 1.7	17.3 ± 0.6	3.1 ± 0.1
Section B2 (around 135 ka)												
VU38-01	150/360	5	−8	60	0.20	0.74	1.59	0.976	32.6 ± 3	26.5 ± 5.3	26.5 ± 5.3	4.7 ± 0.9
VU38-03	150/400	6	−12	53	0.18	0.68	0.84	0.933	20.0 ± 3	26.5 ± 5.3	26.5 ± 5.3	4.7 ± 0.9
VU38-03	150/360	5	−12	52	0.18	0.73	1.26	0.970	28.0 ± 3	26.5 ± 5.3	26.5 ± 5.3	4.7 ± 0.9
VU38-06	200/350	4	−8	58	0.09	0.58	0.81	0.994	25.3 ± 2	26.5 ± 5.3	26.5 ± 5.3	4.7 ± 0.9

average age for these flows. Single age values were used for the entire Sections B1 (104 ± 3 ka) and B2 (135 ± 4 ka). Note that for Section A, two separate palaeointensity means are recorded at about 100 ± 3 ka, which each correspond to different recorded field directions.

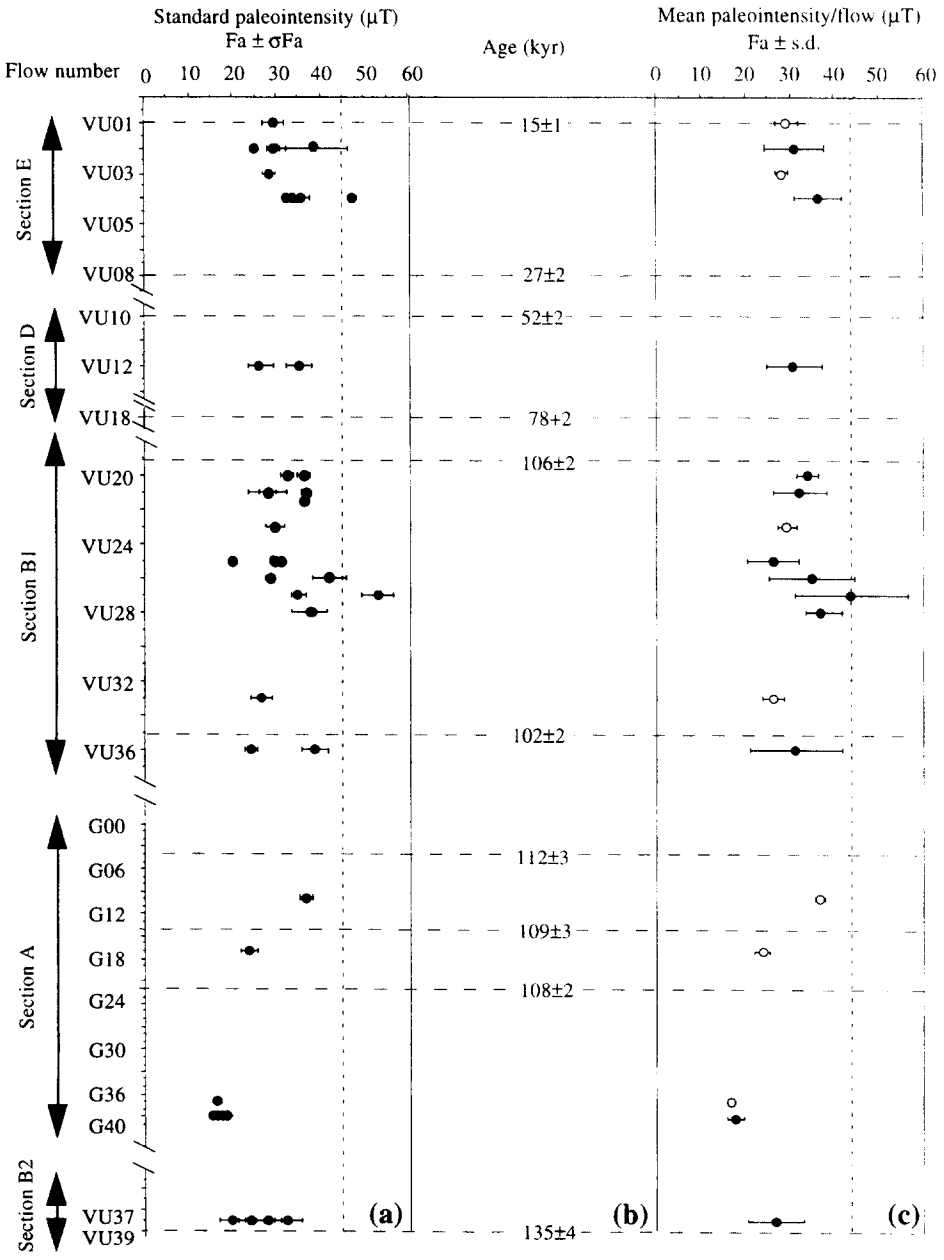


Fig. 10. (a) Palaeointensity versus stratigraphic height with data shown for all accepted samples; (b) K–Ar ages of dated flows; (c) same as (a), but with data averaged from each flow when more than one determination is available. The vertical lines correspond to the present value of the field at the site.

The new results from Vulcano are largely consistent with those from Mt. Etna (Fig. 11). The fact that the two data sets were obtained from different lithologies (the Mt. Etna units are basaltic) indicates that both localities have recorded real geomagnetic field signals. We then combined the results from these two sites to obtain a more accurate description of the geomagnetic field intensity changes in this area. The combined results indicate that the intensity of the geomagnetic field was significantly weaker than today in the Mediterranean area at around 110 ka. Whether this low represents a worldwide characteristic of the dipole field or a particularly large local non-dipole field anomaly can only be ascertained by comparing the Vulcano/Etna results with other determinations from different and geographically widely separated localities.

We then converted the palaeointensity data to their equivalent virtual axial dipole moment (VADM) values which are plotted in Fig. 12 together with VADMs determined from other regions. Many absolute palaeointensity determinations exist for the younger time interval explored at Vulcano. In this period, the dipole moments obtained at Vulcano/Mt. Etna are globally consistent with other determinations from various localities (Fig. 12(a)), suggesting that these flows may have recorded a global (dipole field?) signal during this period.

On the other hand, absolute palaeointensity data are sparse for the interval from 90 to 135 ka, so that comparisons are possible with only a limited data set. Of particular interest is a comparison with recent results reported from the island of La Réunion, Indian Ocean (Chauvin et al., 1991; Raïs et al., 1996), because La Réunion is located in the same middle latitudes as Vulcano, but in the southern hemisphere. The VADM values from La Réunion are consistently higher than those from Vulcano/Mt. Etna by a factor of 3–4 (Fig. 12).

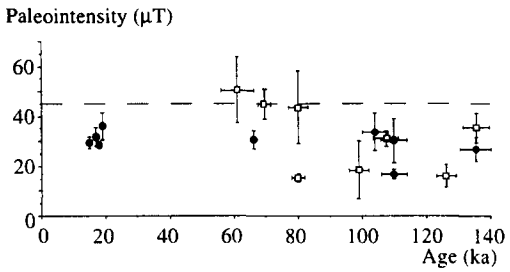


Fig. 11. Changes of the palaeointensity values obtained at Vulcano (solid circles) versus age, compared with the values obtained from Mt. Etna (open squares) (Tric et al., 1994).

Given the paucity of volcanic data for this period, comparison on a wide geographic scale can only be done using sedimentary records of relative geomagnetic field intensity. Although some differences exist between them, the most detailed records (Meynadier et al., 1992; Tauxe and Shackleton, 1994; Weeks et al., 1995; Lehman et al., 1996) generally appear to document a dipole moment strength close to that documented at Vulcano in the period around 100–130 ka, after the records are calibrated with the volcanic data available for the last 50 kyrs. On this basis, the field strength recorded at Vulcano appears to lie within the uncertainties of the estimates for the strength of the dipole field for the period from 100 to 130 ka. The intensity recorded at La Réunion, on the other hand, appears to be significantly higher than the global value. The sedimentary record obtained from the Somali Basin (Meynadier et al., 1992), which is the site closest to La Réunion, also documents a low intensity period. This suggests that the coeval intensity high documented at La Réunion may be spatially limited.

Contemporaneous differences in field strength of the order of those observed between Vulcano and La Réunion may be expected at geographically separated sites because of large contributions from local non-dipole field terms. One would, however, expect these contributions to last for periods of the order of the time constants associated with non-dipole terms, which are generally of the order of a few hundred to maybe 1000 or 2000 years (e.g. Hulot and Le Mouël, 1994; Courtillot and Valet, 1995). We therefore have no ready explanation for this difference, which is recorded over a 25-kyr period, except that this long time constant may reflect core–mantle interactions (Gubbins, 1988).

Differences between Vulcano and La Réunion are also apparent when the directional secular variation is considered. Combining the results from Sections A and B shown in Tables 2 and 3 (with the exception of those corresponding to the large directional deviation recorded in the lower part of Section A at around 110 ka), a value $S = 11.7 \pm 1.0°$ is obtained for the VGP angular dispersion. A corresponding

value $S = 13.9 \pm 1.2°$ is obtained at La Réunion (data from Table 2 in Raïs et al., 1996, where the results from flows 03, 09, 18, 19, 55, 56, 57 and 62 have not been taken into account). This last value is quite consistent with global data (a value of 13.5 is expected) and with the prediction of a model (model G) for palaeosecular variation which correctly fits global data for site latitudes lower than 70° (Merrill and McFadden, 1988; McFadden et al., 1988). The value from Vulcano, on the contrary, is clearly smaller than expected for a site at this latitude (Fig. 13). This result is consistent with the hypothesis of a dominant dipole term at Vulcano/Etna, and a large contribution of non-dipole field at La Réunion. Al-

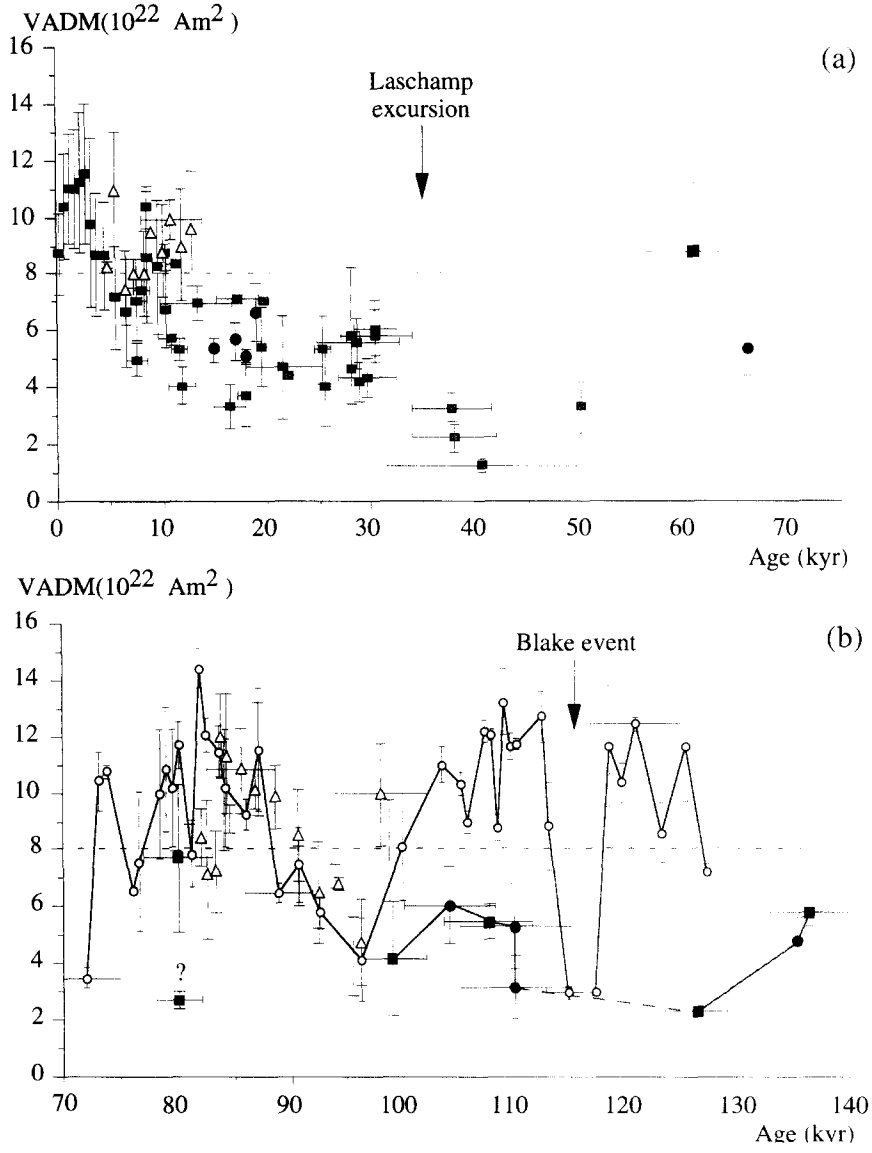


Fig. 12. (a) Comparison of VADMs obtained from Vulcano (solid circles) and Mt. Etna (solid squares) with the worldwide data for the 0–50-ka interval (shaded squares) (see references in Tric et al., 1992) and from La Réunion (Chauvin et al., 1991). (b) Comparison for older ages with data from La Réunion (Chauvin et al., 1991 and Raïs et al., 1996; open triangles and open circles, respectively).

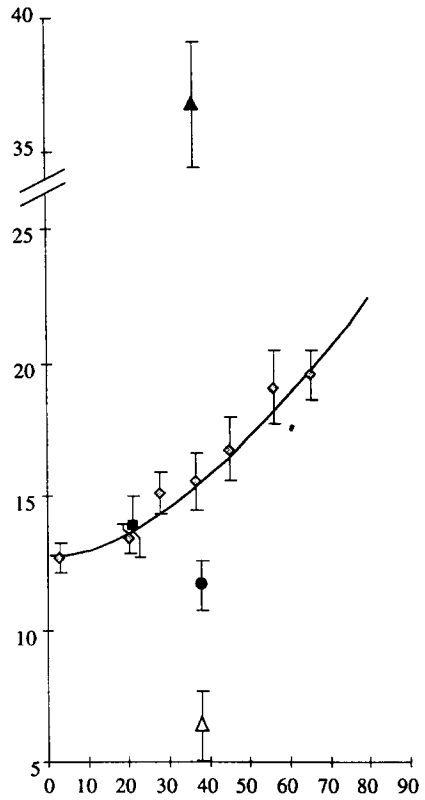


Fig. 13. Comparison of S values from La Réunion (squares), Vulcano non-transitional data (circles) and Vulcano transitional data (triangles) with combined polarity data (diamonds) for the past 5 million years and to model G (solid line) (McFadden et al., 1988). For the transitional data, the full symbol refers to the value obtained with respect to spin axis, and the open symbol with respect to the mean transitional VGP.

ternatively, this small value could result from incomplete sampling of the secular variation, but this appears less realistic to us, because radiometric dates document that a period of at least a few thousand years has been sampled.

The large directional excursion in Section A, with VGPs that cluster over eastern Siberia, coincides in time with the Blake event (Tucholka et al., 1987; Tric et al., 1991). We are therefore tempted to attribute these anomalous directions and intensities to a partial record of this event. Evidence for this event has recently been reported from coeval volcanic rocks in the nearby island of Lipari (Lanza and Zanella, 1991). Furthermore, the record of Tric et al. (1991) obtained from marine sedimentary cores in the nearby Tyrrhenian Sea (39°N, 13°E) contains VGPs from the final part of the Blake event which lie over eastern Siberia, suggesting that the record from Vulcano may correspond to the end of the Blake event. When calculated with respect to the spin axis of the Earth, the angular dispersion for the VGPs corresponding to this excursion (flows G25 to G40 of Table 2) is $S = 36.5 \pm 2.4°$, but this high value clearly arises from the angular distance between the spin axis and the centre of the cluster of transitional VGPs. A much more realistic evaluation of the scatter is obtained by doing the calculation with respect to the mean VGP of the cluster. In this case, the scatter parameter ($S = 6.5 \pm 2.4°$) is well below the values reported above for non-excursional VGPs. As discussed above, this small value could be due to incomplete temporal sampling. Alternatively, one may speculate that the secular variation during the termination of the Blake event was significantly smaller than that observed during non-excursional periods. Additional data are clearly necessary to assess this last point, not only at Vulcano but from other transitional lava flows as well.

As a conclusion, we think that these results contribute to a better knowledge of the geomagnetic field in the past, but they also emphasise the necessity of obtaining data from many more globally distributed localities, before a true picture of the field can be obtained.

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